PETOPIA V2 — Complete Game Design Document

Existing V1 and V2 vision

V1 already includes pet selection, several pet characters, pet growth, feeding, bathing, ball throwing, play interactions, and an adventure world. Preserve these features. V2 extends the existing project with connected 2D Metroidvania exploration, parallax scenery, toy-part collection, Toy Builder, on-foot Pet Racing, Pet Chef, and a broader Game Center. All activities must share persistent pet, currency, and inventory data.

Core loop

Care for pet at home -> explore with pet or choose a minigame -> earn coins, ingredients and toy parts -> cook, craft, customize -> feed and play with pet -> strengthen bond and unlock skills -> explore further. Each mode should affect another mode.

Pet care and real-time growth

Persistent attributes: hunger, thirst, hygiene, energy, happiness, health and bond. Feeding, bathing, sleeping, playing and training use interactive animations. Pets can become mildly unwell when neglected and always recover through accessible care; no permanent death. Time-based development: baby (days 1–7), child (days 8–30), teen (days 31–60), adult (day 61+), adjustable after playtesting. Offline decay is capped; children must not be pressured to check in frequently. Bond affects reactions, tricks, greetings and cooperative abilities.

Characters and abilities

Long-term pet roster: Foxy (hidden objects), Mochi (narrow spaces), Bubbles (water), Ember (fire), Sprout (vines), Frosty (ice), Bolt (machines), Luna (secrets), Poppy (jump), Pebble (shield), Sunny (glide), Cloudy (clouds). MVP: Foxy, Ember and Bubbles. Shared animations: idle, run, jump, fall, eat, drink, wash, sleep, happy, sad, sick, recover, play, ability. Child explorer: idle, run, jump, interact, celebrate, pet and carry. Use consistent pixel density.

Metroidvania connected world

Use a connected graph of 2D rooms rather than only independent levels. Discover locked routes, gain pet abilities, revisit earlier locations, find secrets and toy parts. Keep child-friendly objective markers, maps, checkpoints, hints and shortcuts. Avoid blocking the main story for weeks behind real-time pet growth. Each region combines platforming, exploration, pet-assisted puzzles and optional challenge rooms.

Green Meadows

Home -> Meadow Fields -> River Crossing -> Hidden Cave -> Old Tower -> Forest Gate. First playable vertical slice: walk/jump, pet companion follow, chest containing a toy part, one ability gate, a secret and return route. Visuals: bright sky, distant mountains, rolling hills, village, windmill, trees, river, bridges, grass, flowers and foreground leaves.

Mystic Forest

Enchanted paths, hanging vines, giant trees, glowing mushrooms, crystals, forest village and hidden chambers. Challenges: grow vine bridges, light puzzles, hidden doors and companion abilities. Visuals: deep layered canopy, mist, drifting fireflies, animated plants and glowing crystals.

Robot Factory

Connected factory rooms with conveyors, circuits, switches, gears, power stations, moving platforms and robot workshop. Collect robot heads, arms, wheels and mechanical components. Use industrial silhouettes, distant buildings, pipes, blinking control panels and independently animated gears.

Parallax background architecture

Use separate transparent layers, not a flattened image. Example scroll factors: sky/clouds 0.05, mountains 0.10, hills 0.20, distant structures 0.35, midground trees 0.60, terrain 1.00, foreground vegetation 1.20. Godot 4: Parallax2D nodes with scroll_scale and repeat_size when textures tile seamlessly. Environmental motion (water, fireflies, gears, foliage) is separate animation. Verify seams, pixel snapping, vertical camera movement, contrast and mobile performance.

Environment art production

Create for each world: sky strips, mountain silhouettes, distant vegetation, midground structures, foreground foliage, terrain TileSet (edges/corners/slopes), bridges, doors, chests, platforms, switches, props, animated effects and collision data. Keep source layered files plus exported transparent PNGs. Concept art is inspiration, not production-ready sprites. Establish tile size, base resolution, pivot, palette, naming and sprite scaling before mass production.

Toy collection loop

Adventure chests, secrets, puzzles and minigame rewards grant identifiable toy parts. Inventory records each unique part; blueprint displays progress such as Robot 6/10 and hints for missing parts. Children can assemble already collected pieces immediately and save partial progress. Completed toys become interactive items in the pet room.

Toy Builder mechanics

2D workshop: central blueprint with numbered slots, side tray of collected parts, progress indicator and hint button. Drag a part near its correct slot -> snap, animate and persist; incorrect drops return gently. Provide tap-select/tap-target alternative. Easy 3–5 pieces, normal 6–10, advanced 10–15; no rotation initially. Initial toys: robot 10 parts, airplane 7, car 6. Generic data-driven assembly engine: toy_id, part_id, PNG texture, slot position, snap radius, order/dependencies, installed state, completion reward. Finished toy can be used with pet.

Pet Racing mechanics

Entirely 2D side-view on-foot pet auto-runner, NOT karts. Four pets compete; player times jumps, power-ups and special ability. Magnet attracts coins; speed boost increases speed; shield protects against obstacle; pet ability may double-jump or glide. 45–90 second races. Tracks: Green Meadows Circuit, Mystic Forest Dash, Robot Factory Sprint. MVP: one track, four racers, countdown, rank indicator, coins, finish screen and persistent rewards. Keep AI competitors fair and readable.

Pet Chef mechanics

Choose recipe -> obtain ingredients -> drag/chop -> stir -> timing challenge -> plate -> store cooked meal -> feed pet in home. Simple 2D sprite animations and forgiving touch interactions. Start with three recipes and two preparation mechanics. Meals change hunger and happiness, while preferences influence pet reactions.

Game Center catalog

Other games: Fruit Match, Memory Paws, Catch the Treats, Fishing Friends, Bubble Pop, Pet Hide & Seek, Treasure Dig, Puzzle Pieces, Garden Grow, Color Splash, Pet Bowling, Shape Sorter, Maze Explorer, Dance Party. Each standalone Godot scene uses shared launch, pause, result, reward and save APIs. Target 1–3 minute sessions with child-friendly difficulty and participation rewards.

Godot 4 implementation

Suggested autoloads: GameState, SaveManager, PetManager, InventoryManager, CurrencyManager, QuestManager, SceneRouter. Scene families: Home, PetRoom, AdventureRoom, Player, PetCompanion, GameCenter, PetRacing, PetChef, ToyBuilder. Data definitions: PetSpecies, Item, ToyBlueprint, Recipe, RaceTrack, WorldRoom, Ability, RewardTable. Version save data and migrate V1 safely. Signals: pet_changed, item_collected, toy_completed, minigame_finished, ability_unlocked.

UI and accessibility

Home shows pet mood/needs and clear buttons for Care, Adventure, Game Center and Shop. Adventure shows objective, coins, ability and map. Racing shows position, countdown, pickups and progress. Toy Builder shows blueprint, numbered slots and parts. Chef shows recipe and current action. Support touch, mouse and keyboard, large hit areas, readable pixel fonts, accessible non-drag alternatives, reduced-motion and forgiving feedback.

V2 production roadmap

1 Audit V1 scenes/assets/save data. 2 Implement layered Green Meadows and room transitions. 3 Add toy parts, chests and persistent inventory. 4 Implement generic Toy Builder with one robot. 5 Implement 2D Pet Racing with one track. 6 Implement Pet Chef with three recipes. 7 Extend Mystic Forest and Robot Factory with ability gates. 8 Add Game Center catalog, polish, onboarding, QA and performance tests.

Acceptance criteria

A child can select an existing V1 pet, feed and bathe it, explore Green Meadows with it, find a toy part, return to the workshop, assemble part of a toy, play a 2D race, cook a meal, feed the meal to the pet and relaunch with all progress intact. Pet care, health and growth remain functional. All new systems are modular and do not regress V1.

Developer implementation prompt

Extend the existing Godot 4 PETOPIA V1 project; first inspect its current scenes, autoloads, pet data, save format and animation conventions. Preserve working pet selection, feeding, bathing, ball throwing, growth and adventure. Add connected 2D Metroidvania exploration with Parallax2D multi-layer environments, collectible toy parts, data-driven Toy Builder with partial assembly, on-foot four-pet racing with magnet/boost/shield, and interactive Pet Chef that creates inventory meals. Integrate all modes with the existing pet state, coins and persistence. Deliver testable milestones, sample content and a QA checklist. Do not introduce 3D racing or replace the existing pet simulation.

Concept Art Gallery

01 — a bright colorful 2d pixel cartoon game ui mockup.png

Concept illustration — review design ideas against the written specifications.

02 — a colorful detailed 2d game concept ux infographi.png

Concept illustration — review design ideas against the written specifications.

03 — a colorful detailed game concept ux storyboard co.png

Concept illustration — review design ideas against the written specifications.

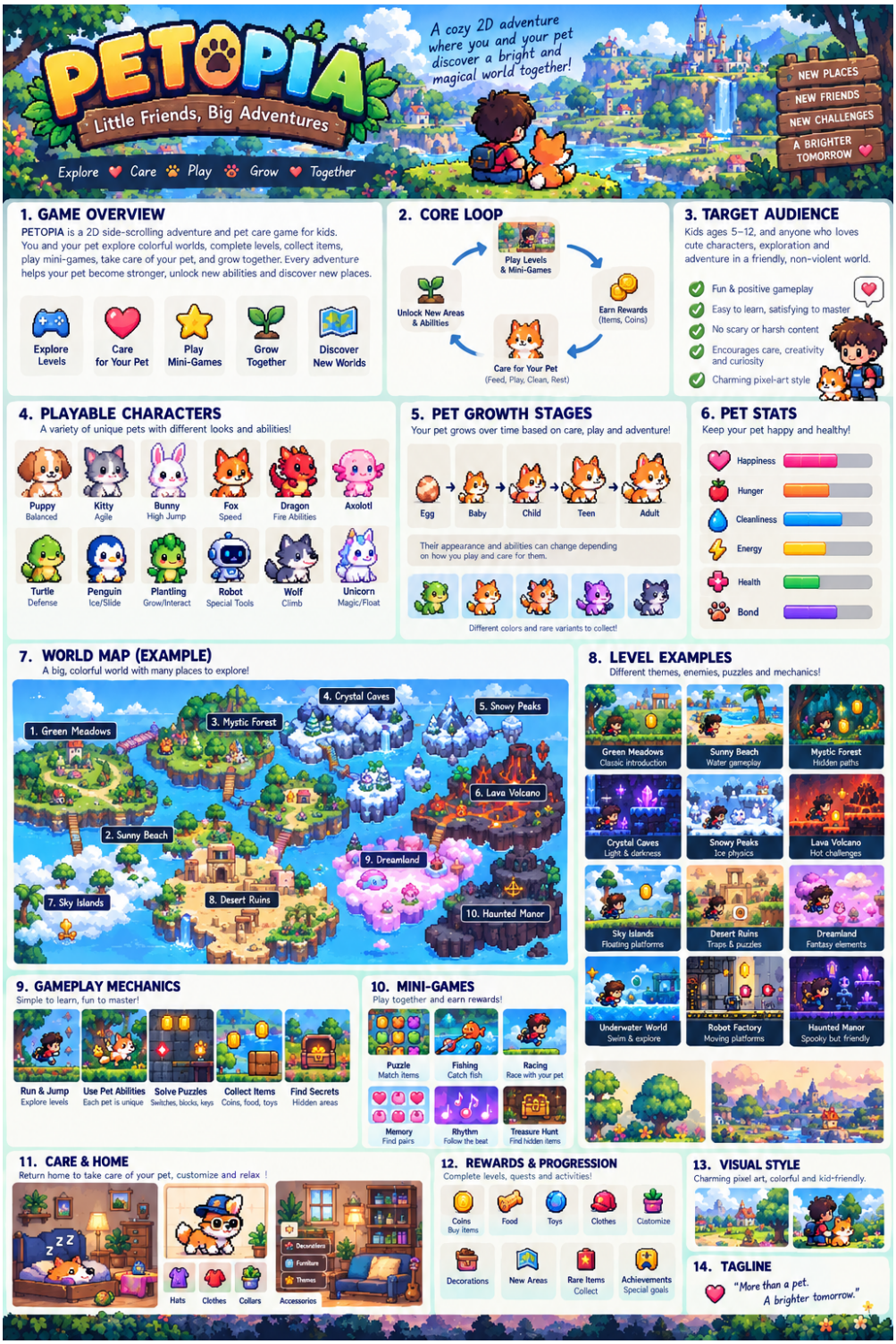
04 — a colorful game concept poster infographic in a pi.png

Concept illustration — review design ideas against the written specifications.

05 — a colorful game concept showcase poster in a cute.png

Concept illustration — review design ideas against the written specifications.

06 — a colorful game design concept document pitch bo.png



Concept illustration — review design ideas against the written specifications.

07 — a colorful high detail game concept poster ui c.png

Concept illustration — review design ideas against the written specifications.

08 — a colorful high detail pixel art infographic port.png

Concept illustration — review design ideas against the written specifications.

09 — a colorful high resolution game concept poster in.png

Concept illustration — review design ideas against the written specifications.

10 — a colorful high resolution pixel art infographic.png

Concept illustration — review design ideas against the written specifications.

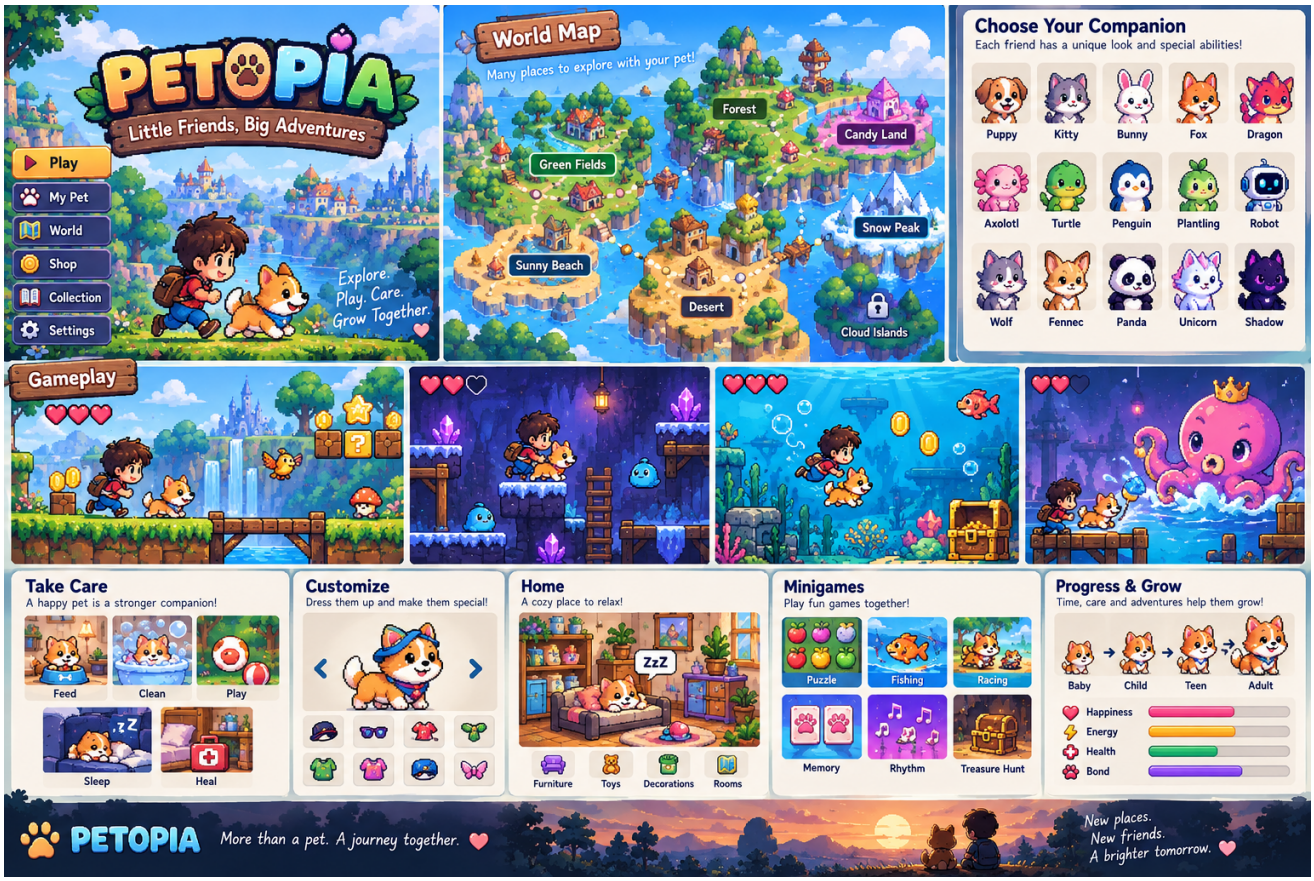
11 — a colorful infographic game ui concept art cart.png

Concept illustration — review design ideas against the written specifications.

12 — a colorful polished 2d cartoon game concept art m.png

Concept illustration — review design ideas against the written specifications.

13 — a colorful polished 2d pixel art game concept col.png



Concept illustration — review design ideas against the written specifications.

14 — a colorful polished pixel art game concept collag.png

Concept illustration — review design ideas against the written specifications.

15 — a colorful promotional game design poster concep.png

Concept illustration — review design ideas against the written specifications.

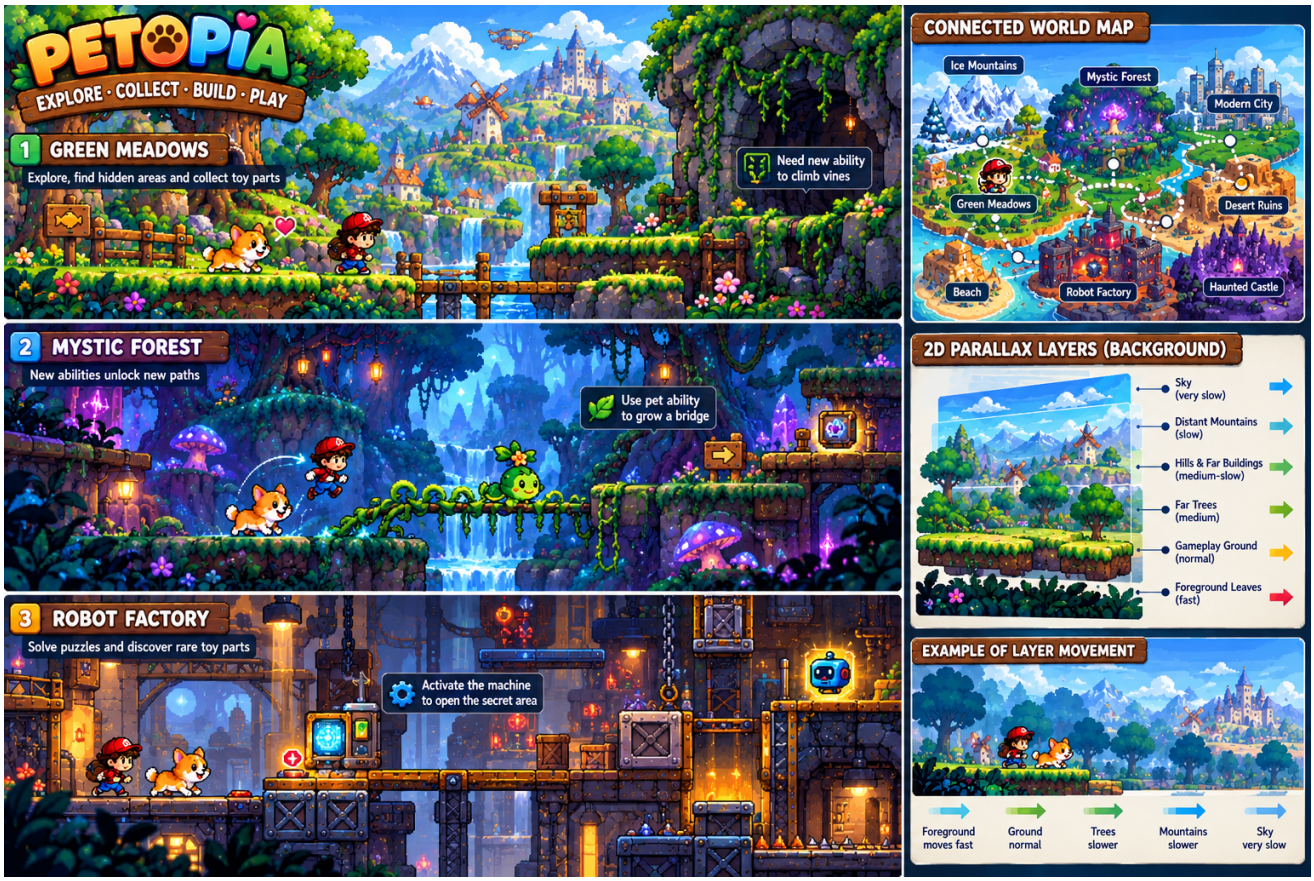
16 — a large colorful game art asset sheet and level b.png

Concept illustration — review design ideas against the written specifications.

17 — a vibrant colorful game concept poster mockup c.png

Concept illustration — review design ideas against the written specifications.

18 — a vibrant detailed game concept poster in a pixel.png

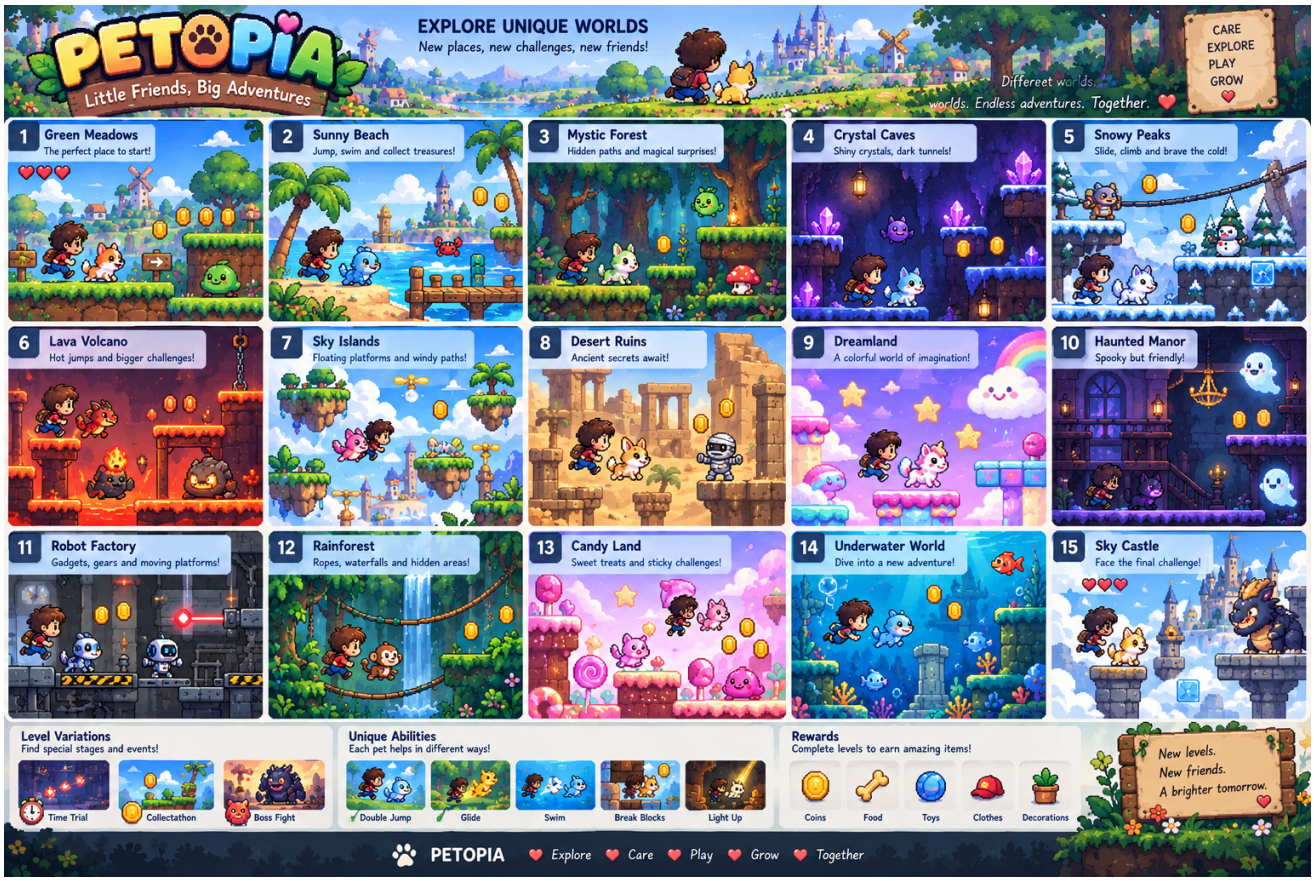


Concept illustration — review design ideas against the written specifications.

19 — a vibrant game concept board collage in a clean c.png

Concept illustration — review design ideas against the written specifications.

20 — a wide colorful game concept poster infographic in.png



Concept illustration — review design ideas against the written specifications.

21 — a wide illustrated game concept poster level des.png

Concept illustration — review design ideas against the written specifications.

22 — wide colorful promotional game concept image in pi.png

Concept illustration — review design ideas against the written specifications.